Question	Answer	Marks	Partial Marks
8(a)(iii)	$(x + 4)(x - 8)$	**M2**	**M1** for $(x + a)(x + b)$ where $ab = -32$ or $a + b = -4$ or for $x(x + 4) - 8(x + 4)$ or $x(x - 8) + 4(x - 8)$
	–4 and 8	**B1**	
8(a)(iv)	3	**2**	**FT** $\frac{12}{their\ 8 - 4}$ **M1** for $\frac{12}{their\ 8 - 4}$ or $\frac{12}{their\ 8} + 1.5$ oe or for answer $\frac{12}{their\ 8}$
8(b)	69.6	**3**	**M2** for $\frac{430 \text{ to } 440}{6 + 0.25}$ or $\frac{440 - 5}{6 \text{ to } 6.5}$ oe or **M1** for 440 + 5 oe or 440 – 5 oe or 6 + 0.25 oe or 6 – 0.25 oe seen
9(a)(i)	(3, 1)	**1**	
9(a)(ii)	$\begin{pmatrix} -10 \\ 15 \end{pmatrix}$	**1**	
9(a)(iii)	3.61 or 3.605 to 3.606	**2**	**M1** for $(-2)^2 + 3^2$ oe
9(b)(i)(a)	$\frac{1}{2}\mathbf{c}$	**1**	
9(b)(i)(b)	$\mathbf{a} + \frac{1}{2}\mathbf{c}$ oe	**1**	FT **a** + *their* (b)(i)(a)

Question	Answer	Marks	Partial Marks
9(b)(ii)(a)	$\overrightarrow{OP} = \frac{1}{3}(2\mathbf{a} + \mathbf{c})$ oe and $\overrightarrow{OQ} = \frac{1}{2}(2\mathbf{a} + \mathbf{c})$ oe OR $\overrightarrow{OP} = \frac{2}{3}(\mathbf{a} + \frac{1}{2}\mathbf{c})$ OR $\overrightarrow{PQ} = \frac{1}{3}(\mathbf{a} + \frac{1}{2}\mathbf{c})$ **and** correct comment e.g. have the same base vector or that they are multiples of one another **and** they share a common point OR e.g. $\overrightarrow{OQ} = 1.5\overrightarrow{OP}$, $2\overrightarrow{PQ} = \overrightarrow{OP}$	**2**	**B1** for $\overrightarrow{OP}$ or $\overrightarrow{PQ}$ factorised or for correct multiplicative statement on relationship without factorised vectors e.g. $\overrightarrow{OQ} = 1.5\overrightarrow{OP}$, $\frac{2}{3}\overrightarrow{OQ} = \overrightarrow{OP}$, $2\overrightarrow{PQ} = \overrightarrow{OP}$, $1.5\left(\frac{2}{3}\mathbf{a} + \frac{1}{3}\mathbf{c}\right) = \mathbf{a} + \frac{1}{2}\mathbf{c}$
9(b)(ii)(b)	1.5 oe	**1**	
10(a)	(2, –10) and (–2, 22)	**5**	**B2** for $3x^2 - 12$ isw or **B1** for $3x^2 + k$ or $px^2 - 12$ ($p \neq 0$) or for $3x^2 - 12 + 6$ isw **M1** for setting *their* derivative = 0 or $\frac{dy}{dx} = 0$ **B1** for $x = \pm 2$ or for one correct coordinate pair
10(b)	(2, –10) minimum with correct reason or sketch (–2, 22) maximum with correct reason or sketch	**3**	**B2** for 1 correct with correct reasoning or **B2FT** for correct evaluation with correct 2nd derivative for both of *their* different x values or **M1** for showing [2nd derivative =] $6x$ or gradients for one value on either side of one correct stationary point or for reasonable sketch of cubic
11(a)(i)	10.2	**2**	**M1** for $\frac{YZ}{13.6} = \frac{15.3}{20.4}$ oe or better